📂 File & Directory Commands

🔹 `ls` – List files

Use: Show directory contents
Syntax:

```
ls [options] [path]
```

Example: `ls -l`

🔹 `cd` – Change directory

Use: Navigate folders

```
cd [directory]
```

Example: `cd /home`

🔹 `pwd` – Print working directory

Use: Show current path

```
pwd
```

🔹 `mkdir` – Create directory

```
mkdir dirname
```

🔹 `rmdir` – Remove empty directory

```
rmdir dirname
```

🔹 `rm` — Delete files/directories

```
rm file
rm -r folder
```

📄 File Handling Commands

🔹 `cp` — Copy files

```
cp source destination
```

🔹 `mv` — Move/rename

```
mv old new
```

🔹 `cat` — View file

```
cat file.txt
```

🔹 `tac` — Reverse file content

```
tac file.txt
```

🔹 `more` — View page by page

```
more file.txt
```

🔹 `head` – First lines

```
head -n 5 file.txt
```

🔹 `tail` – Last lines

```
tail -n 5 file.txt
```

🔹 `touch` – Create empty file

```
touch file.txt
```

🔍 Search & Compare

🔹 `grep` – Search text

```
grep "word" file.txt
```

🔹 `find` – Search files

```
find /path -name file.txt
```

🔹 `locate` – Fast search

```
locate file.txt
```

🔹 `whereis` – Locate binary/source

```
whereis ls
```

🔹 `diff` — Compare files

```
diff file1 file2
```

🔹 `file` — File type

```
file filename
```

📦 Compression Commands

🔹 `gzip` / `gunzip`

```
gzip file
gunzip file.gz
```

🔹 `zip` / `unzip`

```
zip file.zip file
unzip file.zip
```

🔹 `tar` — Archive

```
tar -cvf file.tar dir
tar -xvf file.tar
```

🔹 `zcat` — View compressed file

```
zcat file.gz
```

🔗 Links

- `ln` — Hard link

```
ln file linkname
```

- `ln -s` — Soft link

```
ln -s file linkname
```

📊 Text Processing

- `sort`

```
sort file.txt
```

- `wc` — Count lines/words

```
wc file.txt
```

🖨️ Printing

- `lpr` — Print file

```
lpr file.txt
```

📅 Date & Time

- `date`

```
date
```

- `cal` — Calendar

```
cal
```

🧮 Calculator

- `bc`

```
bc
```

- `bc -l` (advanced math)

```
bc -l
```

👤 User & System Info

- `who` — Logged users

```
who
```

🔹 `whoami`

```
whoami
```

🔹 `w` — Active users

```
w
```

🔹 `finger`

```
finger username
```

📖 Help Commands

🔹 `man` — Manual

```
man ls
```

🔹 `whatis` — Short description

```
whatis ls
```

⚙️ Shell Commands

🔹 `echo`

```
echo "Hello"
```

- `alias`

```
alias ll='ls -l'
```

- `unalias`

```
unalias ll
```

⚡ Process & Job Control

- `ps` — Process status

```
ps
```

- `jobs` — Background jobs

```
jobs
```